

Manthan Sontakke

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate specializing in embedded firmware development and hardware-software integration, with applied experience in AI/ML-driven signal processing and computer vision. Builds IoT and autonomous systems end-to-end from ESP32/Arduino firmware to CNN model deployment.

EDUCATION

Vellore Institute of Technology, Bhopal

Aug 2023 – May 2027

B.Tech, Electronics and Communication Engineering (AI & Cybernetics)

CGPA: 8.66/10

TECHNICAL SKILLS

- **Programming Languages:** Embedded C, C++, Python
- **CS Fundamentals:** Computer Networks, SQL
- **Cloud Services Concepts:** AWS (EC2, S3, IAM, Lambda)
- **Embedded Systems & Hardware:** ESP32, Real-Time Systems, RTOS Concepts, MOSFET, BJT, Op-Amps, Sensor Interfacing, Oscilloscopes, Multimeters, Logic Analyzers
- **Communication Protocols:** UART, SPI, I2C, PWM, Wi-Fi, Bluetooth/BLE
- **AI/ML & Signal Processing:** TensorFlow, CNN, Digital Signal Processing (DSP), FastAPI, Computer Vision, NumPy
- **Tools & Platforms:** Arduino IDE, MATLAB, LTspice, Streamlit, AirSim, Git, ThingsBoard

PROJECTS

AI-Powered RF Spectrum Intelligence Platform — GitHub — *TensorFlow, CNN, DSP*

Jul 2026 – Present

- Designed an AI-based platform for automatic RF communication signal generation, analysis, and classification, supporting 5 modulation schemes (BPSK, QPSK, 16-QAM, OFDM, Multi-Tone).
- Architected a DSP pipeline (AWGN noise modeling, low-pass filtering, FFT, spectrogram generation) that reduced manual signal-analysis effort by 80%.
- Trained a CNN on spectrogram-based features achieving over 90% validation accuracy, deployed via an interactive Streamlit dashboard with real-time FFT visualization and confidence-score prediction.

Autonomous Drone Logistics & Swarm Ground Control Station — GitHub — *AirSim, YOLO*

Jun 2026 – Jul 2026

- Created a Python-based Ground Control Station for autonomous drone delivery and swarm simulation on Microsoft AirSim/Unreal Engine, translating 3-axis (X, Y, Z) simulated telemetry into real-world GPS coordinates for live GIS tracking.
- Implemented a 4-stage autonomous dispatch pipeline (route calculation, takeoff, cruise-altitude navigation, landing) for point-and-click deployment, with thread-safe real-time telemetry pinging.
- Engineered 2 automated scripts for swarm data harvesting and multi-angle image collection, and developed a YOLO-based object detection training pipeline for onboard computer vision.

QUENCH - IoT-Based Contactless Smart Water Dispenser — GitHub — *ESP32, Sensors*

Feb 2025 – Apr 2025

- Engineered system/board design for a contactless IoT water dispensing system integrating ESP32, a water flow sensor, managing power design and component integration.
- Programmed embedded firmware in C/C++ to automate dispensing logic and real-time payment verification, reducing manual user intervention by 70%.
- Achieved a dispensing response time under 2 seconds and 95% flow-sensor measurement accuracy for volumes up to 1500 ml through precise sensor calibration and system validation.

EXPERIENCE

Embedded System Design Intern

Jun 17, 2026 – Aug 2, 2026

Maven Silicon

- Completed a structured embedded systems course covering digital design and Arduino Uno-based embedded system fundamentals, brushing up on core concepts in microcontroller architecture, GPIO, and peripheral interfacing.
- Built an Arduino Uno-based object detection system for visually impaired users, executing system-level unit and integration tests on sensor input and control logic.

AI/ML Intern

May 20, 2026 – July 31, 2026

MPOnline Limited

- Completed a 6-module curriculum covering Python, data preprocessing, and supervised/unsupervised machine learning, applying feature engineering and data visualization techniques to real-world datasets.
- Trained deep learning models using TensorFlow and PyTorch for Computer Vision and NLP tasks across an 11-week program, and deployed an end-to-end AI capstone project via Flask/FastAPI following MLOps practices including Git-based model versioning and pipelines.

CERTIFICATIONS & EXTRA-CURRICULAR

- **Certifications:** Fundamentals of Digital Design for VLSI Chip Design (L&T EduTech / Coursera)
NPTEL Certified in Internet of Things (Score: 88/100) — Applied Machine Learning in Python (Score: 88.63/100)
- **Extra-Curricular:** Discipline Head, Marathi Cultural Club
Represented at the State Level Cricket Tournament (Under-17, VCA).